Bihar Engineering University, Patna
B.Tech 8th Semester Examination, 2025

Course: B.Tech. | *Time: 03 Hours*
Code: 100808 | *Subject: Total Quality Management* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[2 x 7 = 14]**

(a) "Quality is defined by the customer" is
(i) An unrealistic definition of quality
(ii) A user-based definition of quality
(iii) A manufacturing-based definition of quality
(iv) A product-based definition of quality

(b) In Six Sigma, a ____ is defined as any process output that does not meet customer specifications
(i) Error
(ii) Cost
(iii) Quality
(iv) Defect

(c) What is the kind of focus put in a company's processes in TQM?
(i) Steady improvement
(ii) Continual improvement
(iii) Passive improvement
(iv) Unsteady improvement

(d) ISO 9000 determines
(i) If the company practices its written procedures
(ii) If vendors are performing well
(iii) Process capability
(iv) Random causes of variation

(e) QFD stands for _____.
(i) Quantity for deployment
(ii) Quality for deployment
(iii) Quality function deployment
(iii) Quality for decision

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(f) Kaizen is a _____ process, the purpose of which goes beyond simple productivity improvement.
(i) Weekly
(ii) Daily
(iii) Monthly
(iv) Annual

(g) In TQM, suppliers are treated as?
(i) Partners
(ii) Employees
(iii) Managers
(iv) Enemies

(h) Which of the following is a dimension of 'product quality'?
(i) Probability
(ii) Machinability
(iii) Durability
(iv) Customer Satisfaction

(i) Which of the following is a principle of TQM?
(i) Process-Centered
(ii) Product-Centered
(iii) External customer focus only
(iv) Internal customer focus only

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(i) DOE identifies

(i) Causes of delayed shipments
(ii) Factors that affect the output
(iii) Traffic congestion points
(iv) Control limits

Q.2 With example, explain six sigma concept and its application to manufacturing sector. **[14]**

Q.3 (a) Briefly explain the major steps in benchmarking. **[7]**
(b) Write about the system of recognition and reward followed in an organisation. **[7]**

Q.4 (a) How are 5S implements in a factory? **[7]**
(b) Explain the seven step plan to establish the TPM in an organization in detail. **[7]**

Q.5 (a) What is quality cost? Explain the techniques used for Quality cost? **[7]**
(b) Explain in detail about Deming Philosophy. **[7]**

Q.6 (a) What is FMEA? Explain the stages of FMEA? **[7]**
(b) Explain the implementation and documentation of Quality System? **[7]**

Q.7 (a) Explain the concepts of Leadership? **[7]**
(b) What are the objectives of Performance measures? Discuss in detail. **[7]**

Q.8 Explain the concept of Taguchi's quality loss function in detail. Give an example. **[14]**

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Q.9 Explain the following **[$3\frac{1}{2}$x4=14]**

(a) Empowerment
(b) Customer satisfaction
(c) Quality
(d) Kaizen

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